

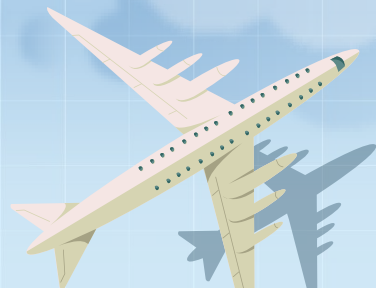
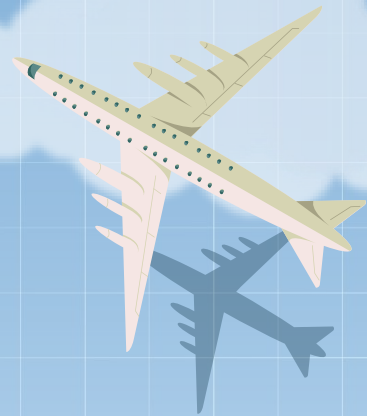
BOEING TEXT MINING PROJECT

Using LLM to Derive Insights from Aircraft
Maintenance Records

CalPoly
Pomona



Boeing



MEET THE ADVISORS



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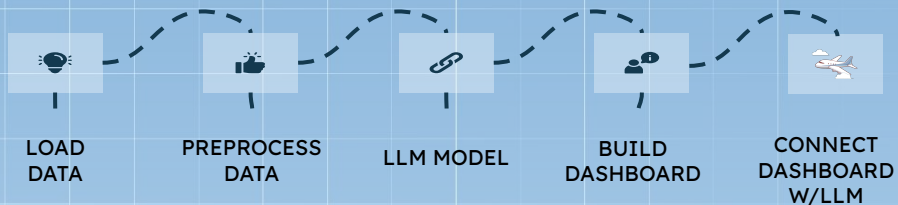


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METHODOLOGY



KEY FEATURES

Objects, Defects, Stringers, Location (Station, Side, Type)

(Structure Corrosion Data)

(Body Station Focused Data)

PROBLEM

Aircraft maintenance databases have a large amount of **freeform** textual information

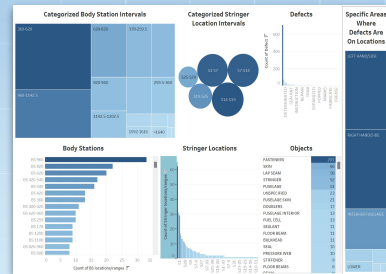


SOLUTION

A Named Entity Recognition (NER) model can label the keywords in each report.



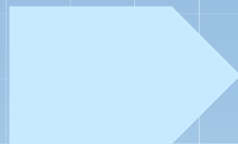
This information will automatically update the dashboard, easily visualizing aircraft defect trends **in real time**.



THE OUTLINE

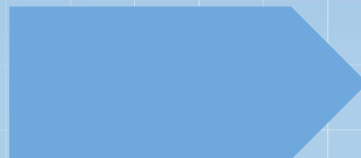
01

DATA PREPROCESSING



02

DATA ANALYSIS



03

NAMED ENTITY RECOGNITION MODEL



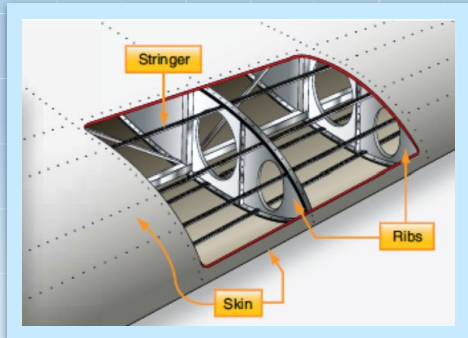


DATA PREPROCESSING

WHAT ARE THE KEY COMPONENTS?

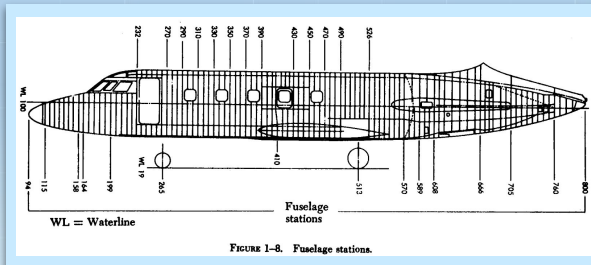
STRINGERS

- **Purpose:** Provides strength and rigidity to the aircraft's skin.
- **Description:** Slender structural components, similar to rods.



BODY STATION

- Body frames of the aircraft
- Organized from nose to tail of the aircraft



LOCATION STATION

- References to its assigned number of body station/stringer

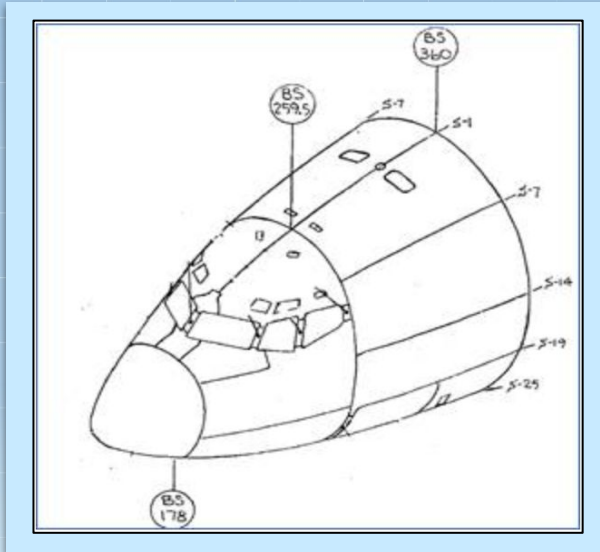
Location Station Begin

3
178
179
189
190
195
196
200
203
204
211
212
219

Location Station End

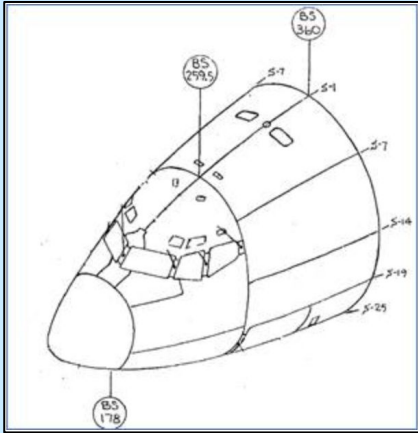
0
178
203
208
210
211
215
219
220
227
230

**BODY STATION LOCATIONS ARE MAPPED AND
GROUPED TO ASSOCIATED RANGE
INTERVALS OR “BUCKETS”**

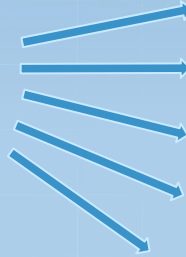


BS
<178
178-259.5
259.5-360
360-620
620-820
820-960
960-1142.5
1142.5-1202.5
1202.5-1322.5
1322.5-1440
1440-1462.5
1462.5-1505.8
1505.8-1592
1592-1616
1616-1640
>1640

BODY STATIONS & STRINGER GROUPINGS



**Body
Station
Bucket**



Stringer

S1-S7

S7-S14

S14-S19

S19-S25

S25-S29

=

BS

Stringer

178-259.5

S1-S7

178-259.5

S7-S14

178-259.5

S14-S19

178-259.5

S19-S25

178-259.5

S25-S29

259.5-360

S1-S7

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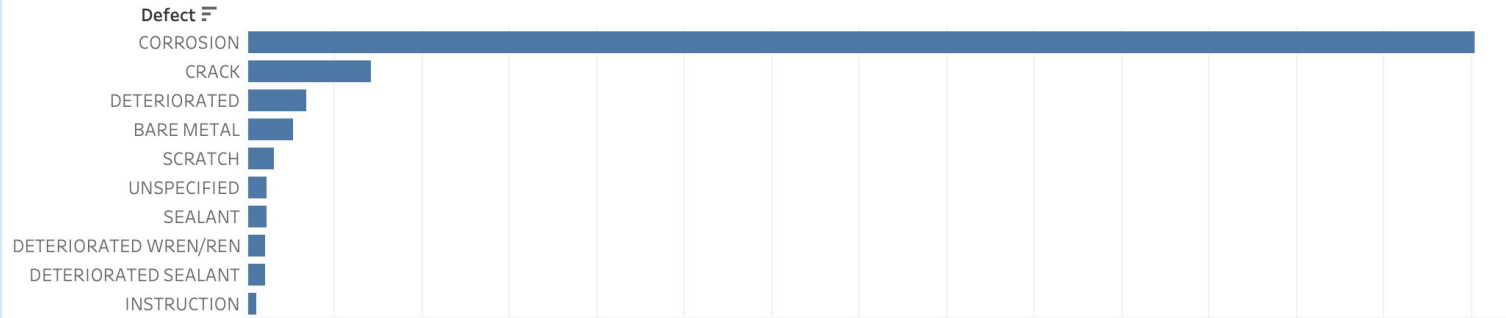
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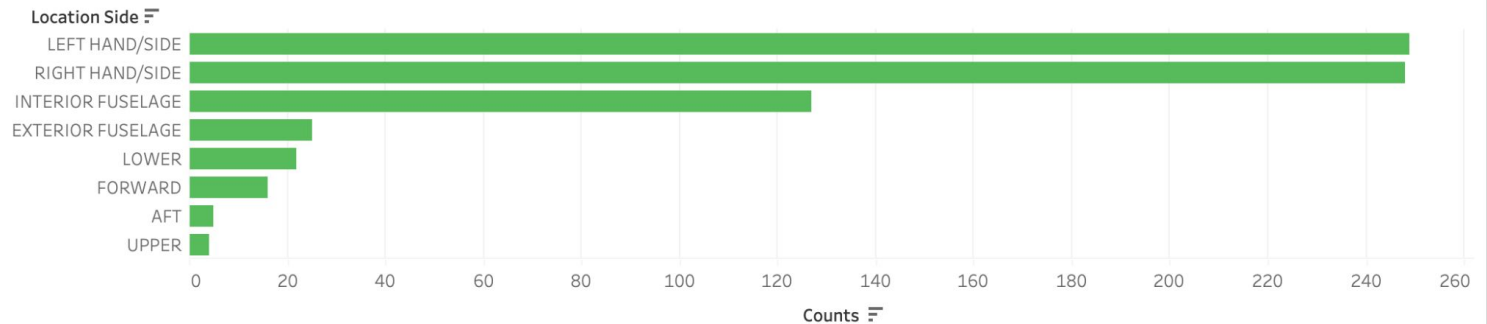


DATA ANALYSIS

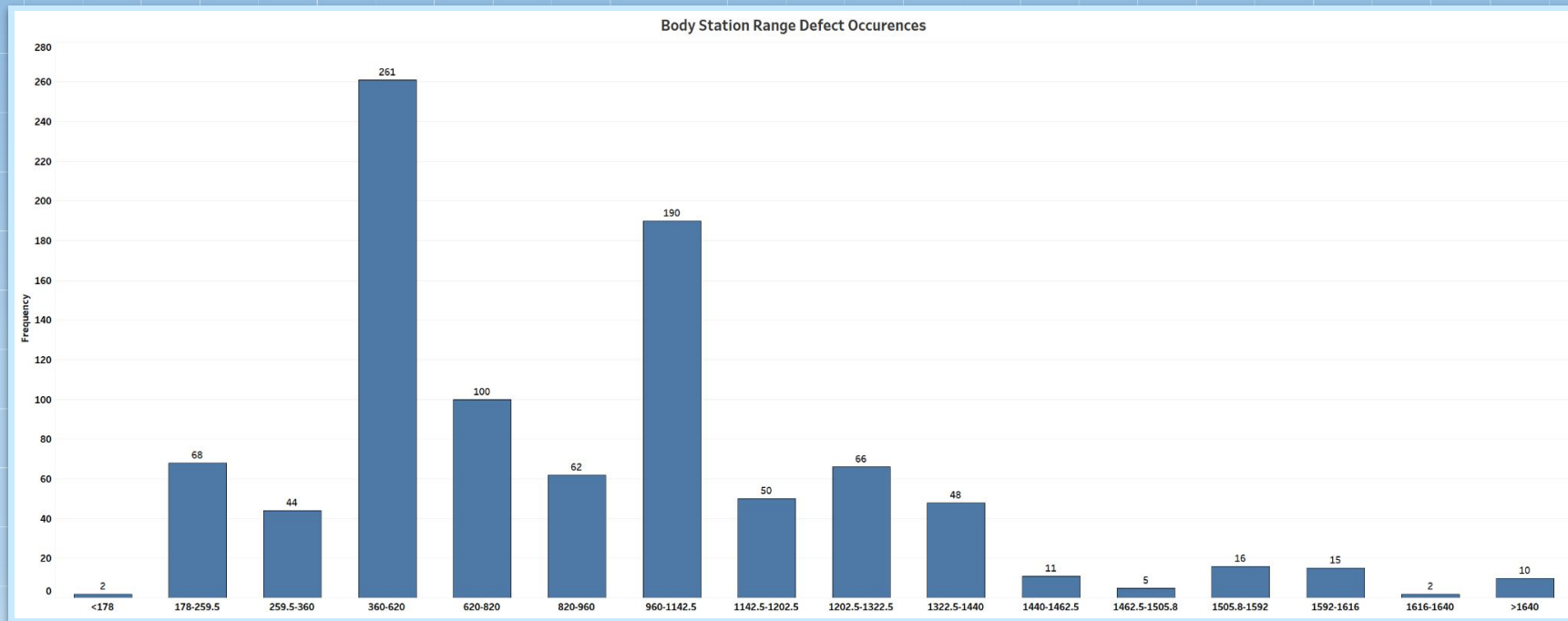
Top Defects



Top Location Sides

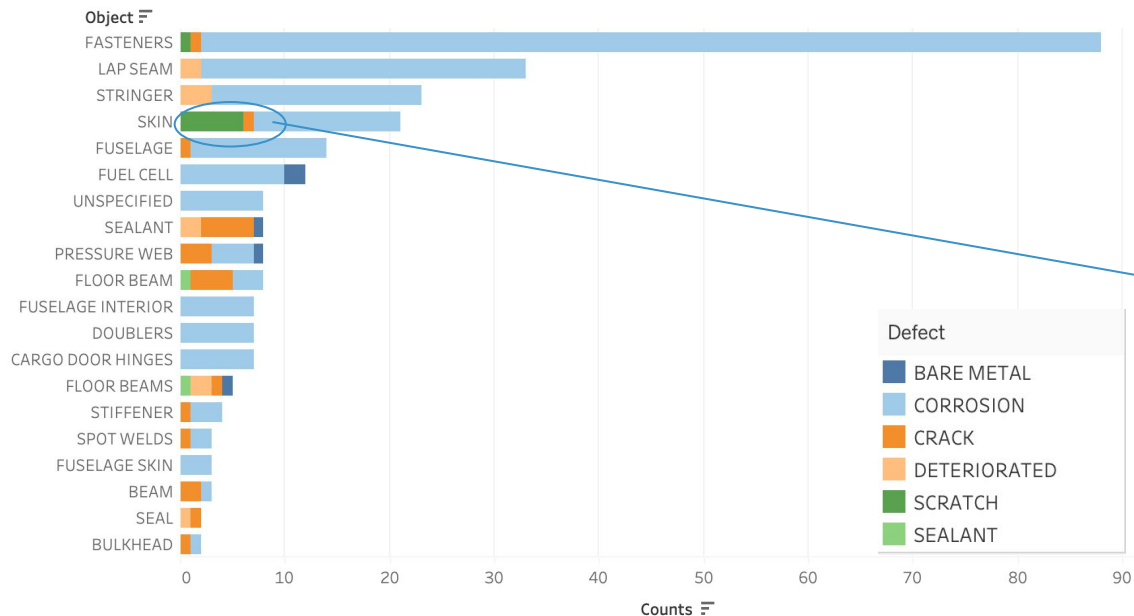


NUMBER OF OCCURRENCES FOR EACH BODY RANGE



Ranges **360-620** and **960-1142.5** tend to have the most occurrences.

NUMBER OF OCCURRENCES (OBJECTS AND DEFECTS) FOR BS RANGES 360-620 AND 960-1142.5

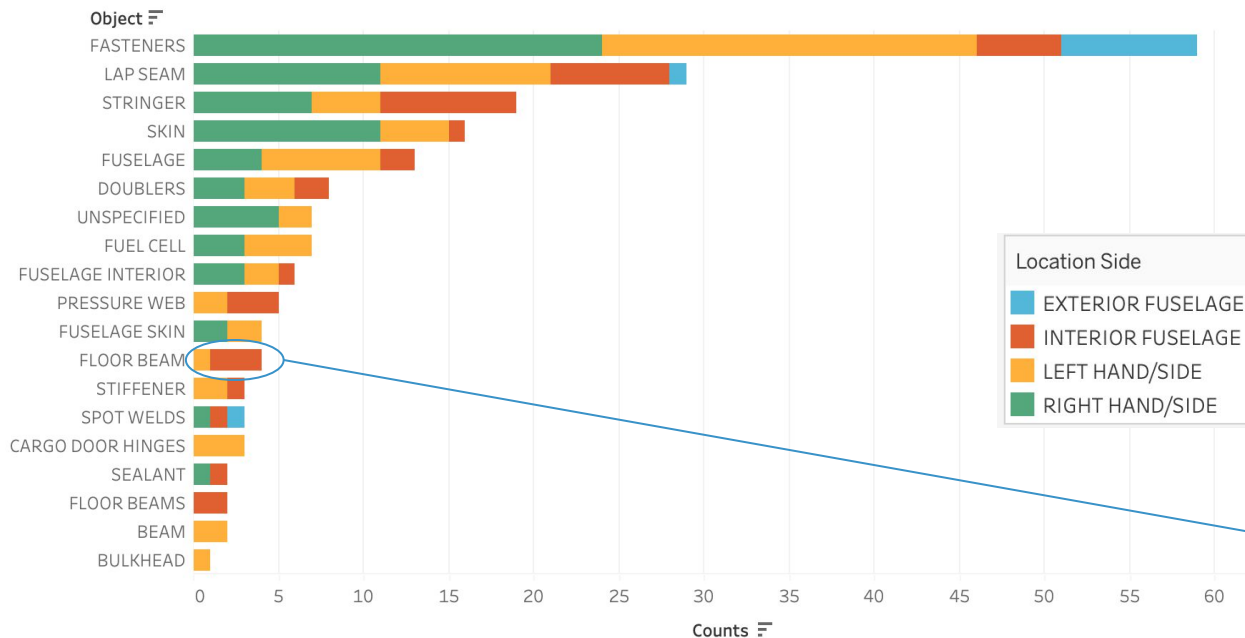


Corrosion occurs frequently on
most objects

Majority of scratches occur on the
skin

Bare metal happens more on
aircraft structural parts

NUMBER OF OCCURRENCES FOR OBJECT LOCATION SIDE



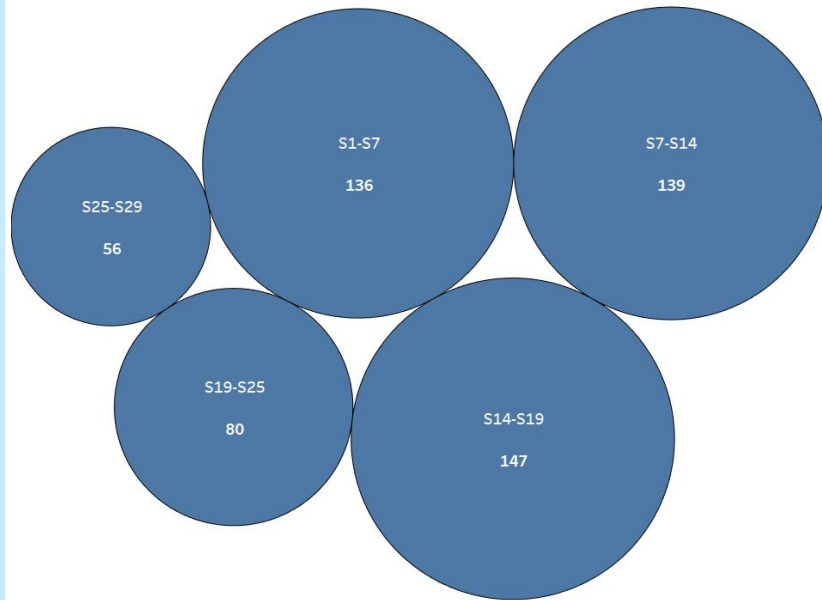
Most exterior fuselage occurrences occur on the fasteners

Most occurrences on Skin are on the Right Hand Side

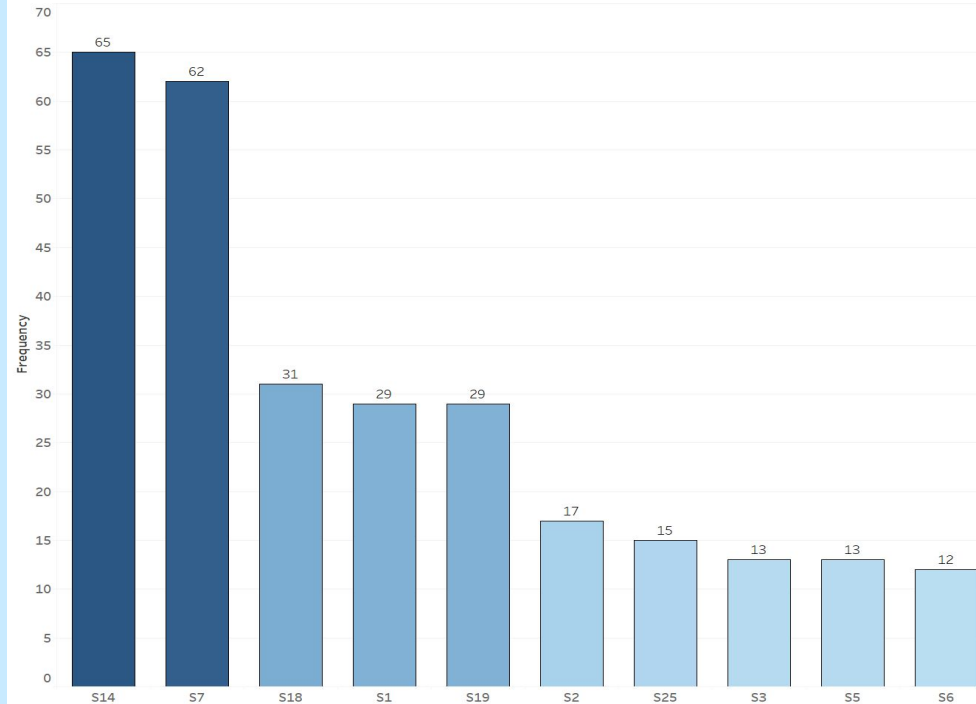
Most occurrences on floor beams occur in the interior fuselage

STRINGERS WITH MOST OCCURENCES

STRINGER RANGE INTERVALS



TOP STRINGER LOCATIONS

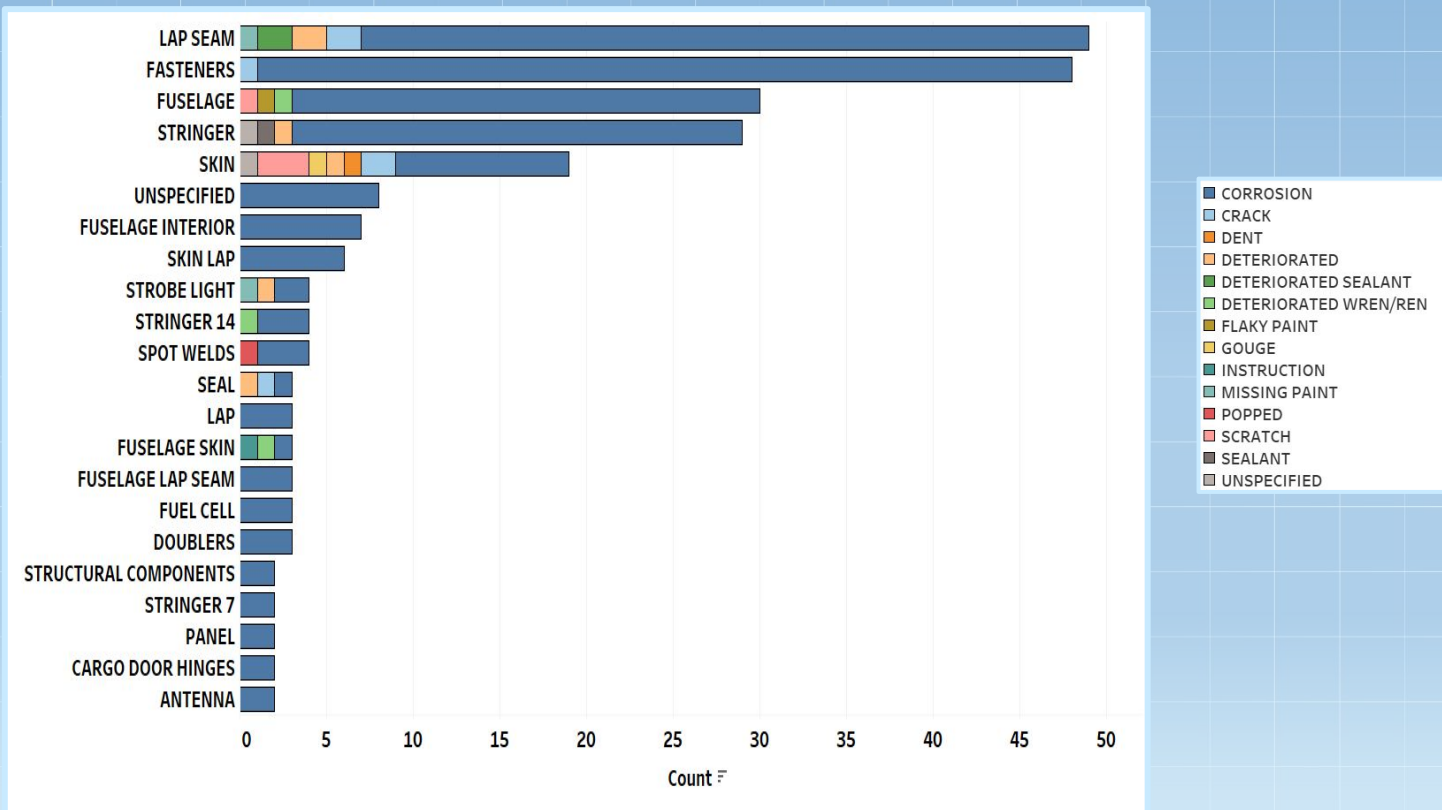


OCCURRENCES OF OBJECTS & DEFECTS IN TOP STRINGER LOCATIONS

For Stringers, **Lap Seam** has slightly more occurrences than Fasteners

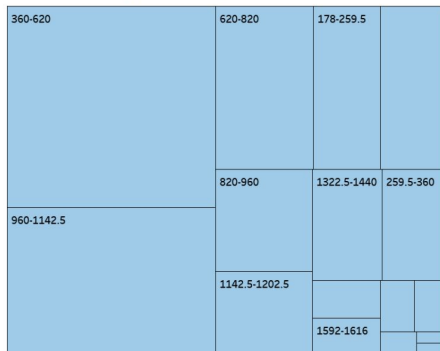
Corrosion is the most
occurring defect
(again)

Majority of the objects
with records
associated with
Stringers seem to be
**aircraft structural
parts**



INTERACTIVE DASHBOARD

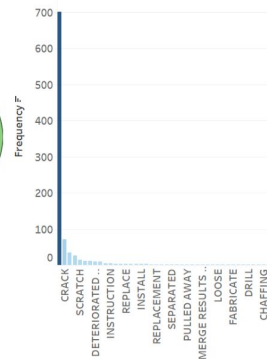
Categorized Body Station Intervals



Categorized Stringer Location Intervals



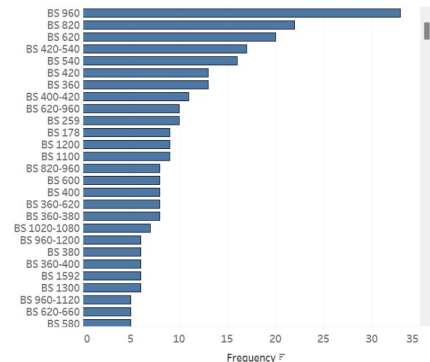
Defects



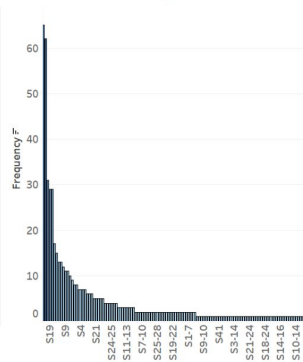
Specific Sides Where Defects Reside On Body Stations



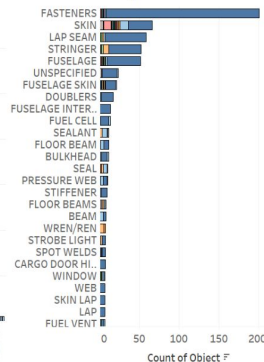
Body Stations



Stringers



Objects



KEY INSIGHTS

- Most cases of corrosion happens equally likely on **left and right hand side**
- Most Body Station Range Occurrences were on **360 - 620** and **960 - 1142.5**
- Majority of **scratches** occur on the **skin** of the aircraft
- Corrosion defects also occur within the **interior and exterior fuselages**

These insights are based on a **limited dataset**. There may be biases!

LARGE LANGUAGE MODEL - BERT (NAMED ENTITY RECOGNITION)



WHAT IS BERT ?



Google

Developed
by Google



Pre-trained on
large corpus of
text



Think of it
as a
smaller
ChatGPT



Understands
meaning of words
by looking at
context



Can use the model
for different types
of NLP tasks
(NER, Sentiment
Analysis)

NAMED ENTITY RECOGNITION

Locates and classifies named entities from unstructured text into predefined categories

REPORT

REF 1D 165 9 7
there is a sign of
corRosion in the
R H 129 450 WS in
STR 198 201 on
skIn Panel



LABELS

OBJECT

SKIN PANEL

DEFECT

CORROSION

LOCATION-TYPE

WS

LOCATION-SIDE

R H

LOCATION-START

129

LOCATION-END

450

STRINGER-START

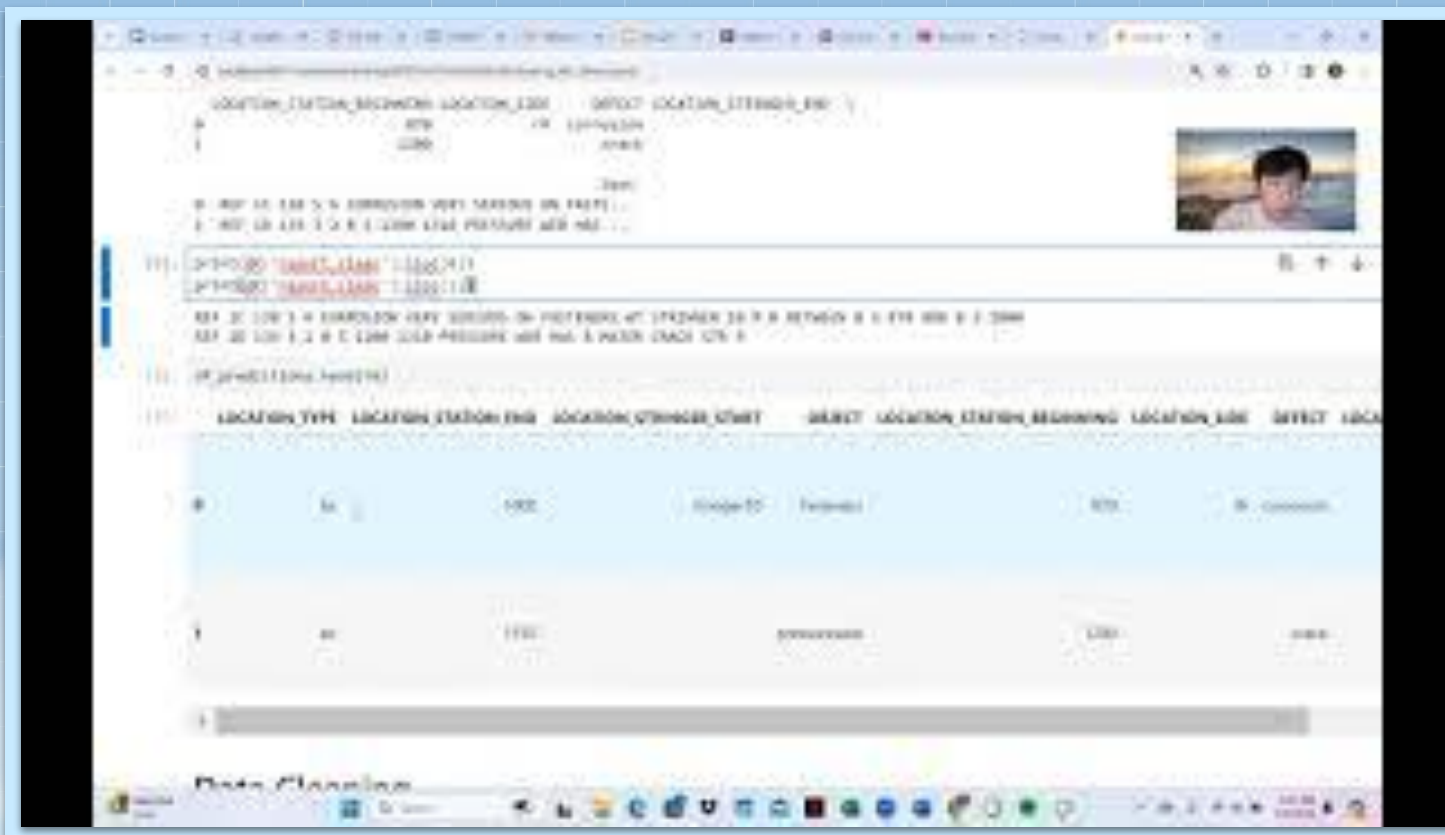
198

STRINGER-END

201

LIVE DEMO

This demo will demonstrate how the end-user can add new data into a Jupyter Notebook, have it generate labels for all of the new reports, and be saved as a CSV file.



REFLECTIONS

ACHIEVEMENTS

- LABELED WORDS INTO CATEGORIES
 - CORRECT 81% OF THE TIME
- CREATED A TRAINED MODEL
 - RESULTS: UPDATES THE DASHBOARD

FUTURE STEPS

- TRAIN ON MORE DATA TO IMPROVE ACCURACY
- USE A GPU
- TRAIN A LLM THAT SPECIALIZES IN AVIATION

CHALLENGES

- MANUALLY LABELING EACH ENTITY FOR MODEL TRAINING
 - TIME CONSUMING AND LIMITING
- USED A CPU
 - TOOK HOURS TO TRAIN DATA

THANK YOU

